Upcoming + Reminders

Find all links on the **course website:** cogs9.github.io/fa26

📚 **Project:** *Checkpoint 1* DUE: Wednesday)

All due at 11:59pm

📚 **Read:** *Reading(s) 4, Data Visualization*

- Reading quiz 4 (DUE: Wednesday Week 6)

Guest Lecture — Week 8

- Participation is worth 1 point
  - Fill out a Google Form of your takeaways
  - Interactive Q&A!
- **Microsoft AI** generates tools like Github Copilot
- Jolene also worked (with me) at **Reddit**

[recorded] Week 8

AI and Industry Data Science



A. Jolene Mork

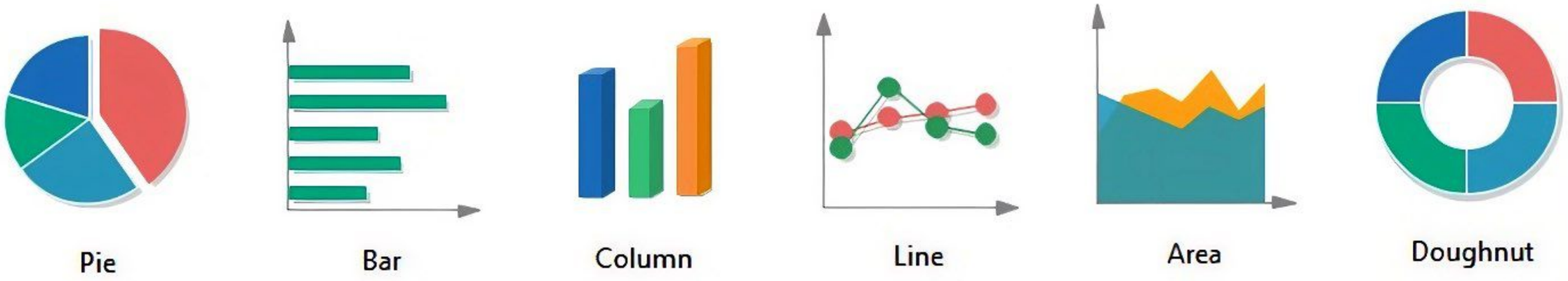
cogs 9 | Week 5

Exploratory data analysis (EDA)

Brendan Tomoschuk

Last time: Data Visualization

- Visualizations shape interpretation by making some patterns more visible or obvious
- Visual encodings determine how easily and accurately people understand data
- Let your question and data determine your choice of visualization



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course code:
WSTAPX

Which visual encoding out of the following is generally most precise for comparing quantitative values?

(A) Color

(B) Angle

(C) Length

(D) Area

(E) I'm not sure / I'm confused

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course code:
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Which visual encoding out of the following is generally most precise for comparing quantitative values?

(A) Color

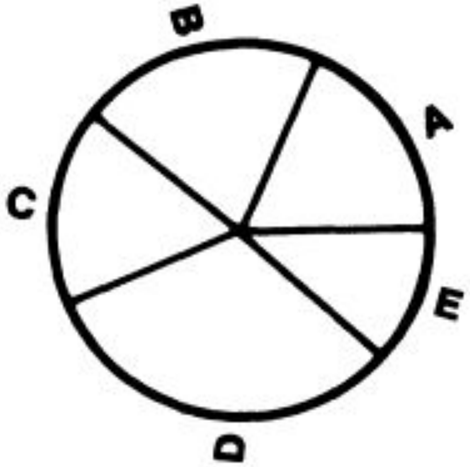
(B) Angle

(C) Length

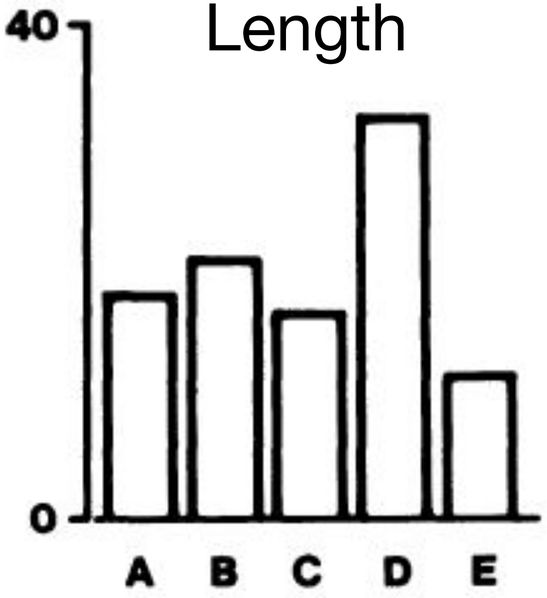
(D) Area

(E) I'm not sure / I'm confused

Area

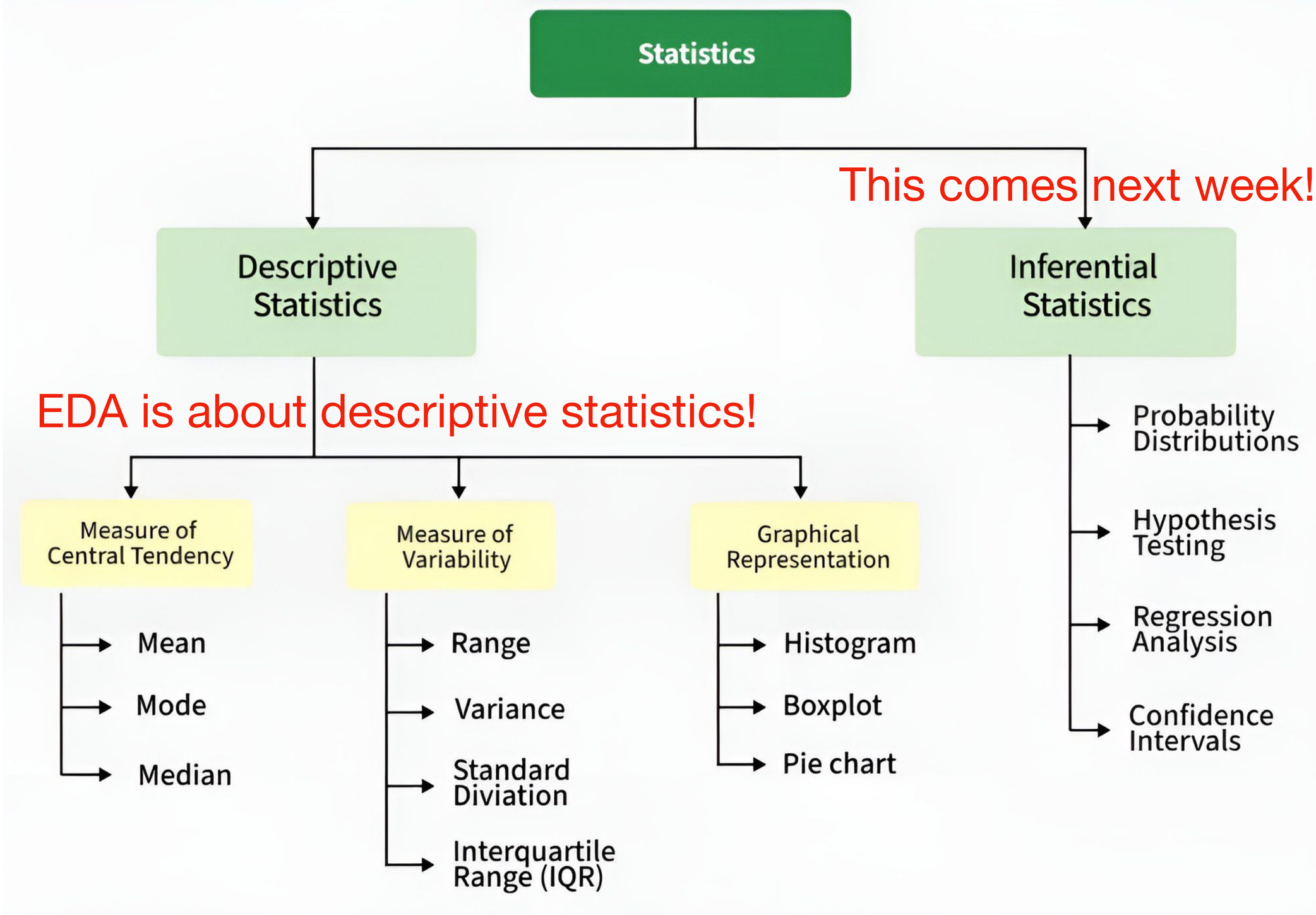


Length



Learning Goals

1. Understand the purpose of exploratory data analysis (EDA)
2. Distinguish different types of EDA and their goals (univariate, bivariate, multi-variate)
3. Explain the basic exploratory data analysis you would carry out on a dataset


Statistics
This comes next week!
Descriptive Statistics
Inferential Statistics
EDA is about descriptive statistics!
Measure of Central Tendency
Measure of Variability
Graphical Representation
Mean
Mode
Median
Range
Variance
Standard Diviation
Interquartile Range (IQR)
Histogram
Boxplot
Pie chart
Probability Distributions
Hypothesis Testing
Regression Analysis
Confidence Intervals

Why do we do EDA?

Main purpose: To understand your dataset before further analysis and modeling!

- Understand basic data properties
- Catch mistakes: reveals missing data, coding errors, impossible values, and duplicates
- Discover patterns and structure: shows distributions, outliers, clusters, and relationships you wouldn’t notice in a table
- Guides analysis choices: helps you pick appropriate models/tests and check modeling assumptions
- Generates hypotheses: you may find patterns in data worth investigating

Why do we do EDA?

Main purpose: To understand your dataset before further analysis and modeling!

- **Step 1:** Understand the data
- **Step 2:** Univariate EDA (1 variable)
- **Step 3:** Bivariate EDA (2 variables)
- **Step 4:** Multi-variate EDA (2+ variables)

Exploring the COGS 9 dataset!

Year	Preferred pronouns (optional)	What is your major?	Do you plan to attend lectures?	What is your height in inches?	How old are you?	What's your natural hair color?	How do you describe your race and/or ethnicity? (Select all that apply)	Where were you born?	How many siblings do you have?	How many hours of sleep do you get a night on average?	What is your spotify listening age?	How many instagram followers do you have?	How many people do you follow on instagram?	What's your favorite quarter?	Do you prefer...	Coffee or tea/matcha?	Cats or dogs?	If money wasn't an object, what would you want to be when you grow up?
Third-year	he/him	Data science	Yes	69.0	22.0	Black	Asian	North or Central America	1.0	10.0	24.0	485.0	740.0	Fall	Tues/Thurs classes	Neither	Dogs	Someone who can help people
Second-year	he/him	Data science	Sometimes	65.0	19.0	Black	Asian, Indian	Asia	0.0	6.0	74.0	898.0	1107.0	Spring	Tues/Thurs classes	Coffee	Dogs	A professional billiards player
Second-year	she/her	Other	No	62.0	19.0	Black	Asian	North or Central America	2.0	7.0	NaN	568.0	327.0	Spring	Tues/Thurs classes	Neither	Dogs	An entrupreneur
First-year	he/him	Other	Yes	74.0	19.0	Black	Asian	Asia	1.0	9.0	0.0	76.0	105.0	Fall	Tues/Thurs classes	Tea/Matcha	Cats	capybara
First-year	she/her	Cognitive science	Yes	NaN	18.0	Brown	Asian	Asia	1.0	5.0	NaN	69.0	72.0	Winter	Tues/Thurs classes	Coffee	Cats	A marine biologist
First-year	he/him	Other	Yes	6.0	18.0	Brown	White	North or Central America	0.0	8.0	50.0	160.0	300.0	Fall	Mon/Wed/Fri classes	Neither	Cats	Writer
Second-year	she/her	Cognitive science	Yes	67.0	20.0	Brown	White	North or Central America	1.0	7.0	65.0	378.0	353.0	Spring	Mon/Wed/Fri classes	Coffee	Dogs	Music teacher
First-year	he/him	Data science	Yes	65.0	18.0	Black	Black or African American	North or Central America	1.0	7.0	62.0	115.0	170.0	Fall	Mon/Wed/Fri classes	Coffee	Cats	NaN

Step 1: Understanding the data

Main goal: Understand your dataset and what's in it

- How many observations are there? *n = 278*
- How many variables? *p = 25*
  - How many are quantitative? *p = 7*
  - How many qualitative? *p = 14*

Why is this important?: Basic things like dataset size and variable types *tell you what kinds of conclusions you can draw* and what *plots, analyses, or modeling tools you can use*

Step 1: Understanding the data

Qualitative

Quantitative

- First name
- Last name
- Year
- Preferred pronouns (optional)
- What is your major?
- Do you plan to attend lectures?
- What's your natural hair color?
- How do you describe your race and/or ethnicity? (Select all that apply)
- Where were you born?
- What's your favorite quarter?
- Do you prefer T/Th or M/W/F classes?
- Coffee or tea/matcha?
- Cats or dogs?
- If money wasn't an object, what would you want to be when you grow up?
- What is your height in inches?
- How old are you?
- How many siblings do you have?
- How many hours of sleep do you get a night on average?
- What is your spotify listening age?
- How many instagram followers do you have?
- How many people do you follow on instagram?

Step 1: Understanding missingness

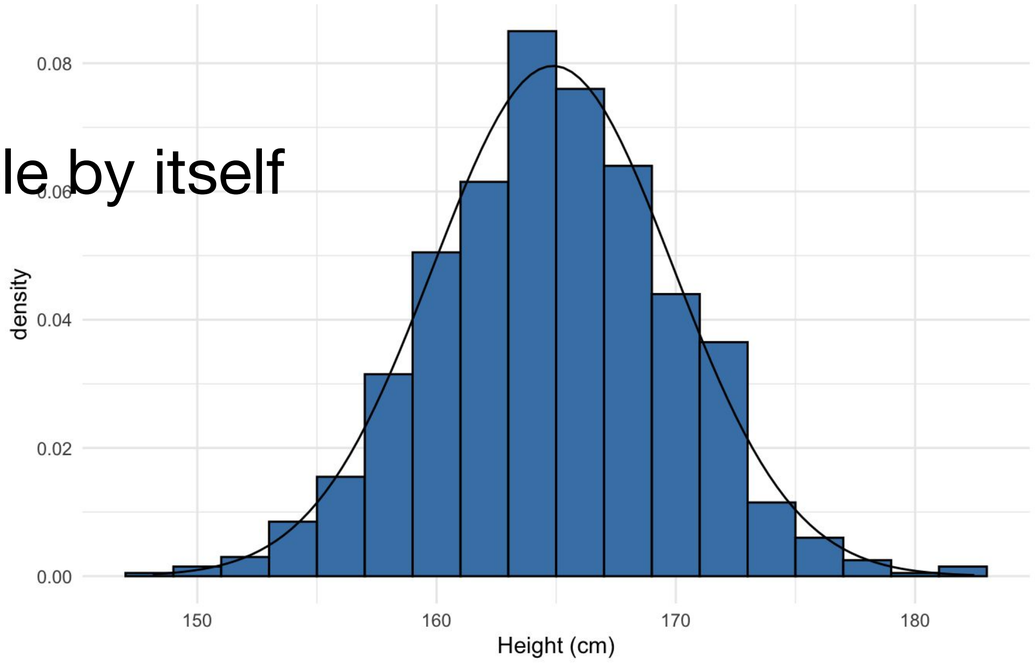
- Are there missing data?
- How much is missing?
- Why are they missing?
- Is the missingness random, or does a *certain group* tend to have more missing data?
  - *e.g. 4th-years answer less survey questions*
- Missingness can bias analysis
- Some models handle missingness better than others!

Variable	# missing	%
Anything else you'd like me to know about you?	194	70.0%
What is your spotify listening age?	122	44.0%
If money wasn't an object, what would you want to be?	74	26.7%
What's something you're really excited to learn or do?	68	24.5%
How many people do you follow on instagram?	54	19.5%
How many instagram followers do you have?	54	19.5%
What is your height in inches?	36	13.0%
What's your favorite quarter?	35	12.6%
Do you prefer T/Th or M/W/F classes?	30	10.8%
How many hours of sleep do you get a night?	25	9.0%
Cats or dogs?	24	8.7%
Coffee or tea/matcha?	22	7.9%
How many siblings do you have?	20	7.2%
How old are you?	19	6.9%
Where were you born?	15	5.4%
How do you describe your race and/or ethnicity?	15	5.4%
What's your natural hair color?	14	5.1%
Preferred pronouns	13	4.7%

Step 2: Univariate EDA

Main goal: Understanding each individual variable by itself

- Descriptive statistics like mean/median/std
- Checking for impossible values and outliers
- Distribution, skew



Why is this important?:

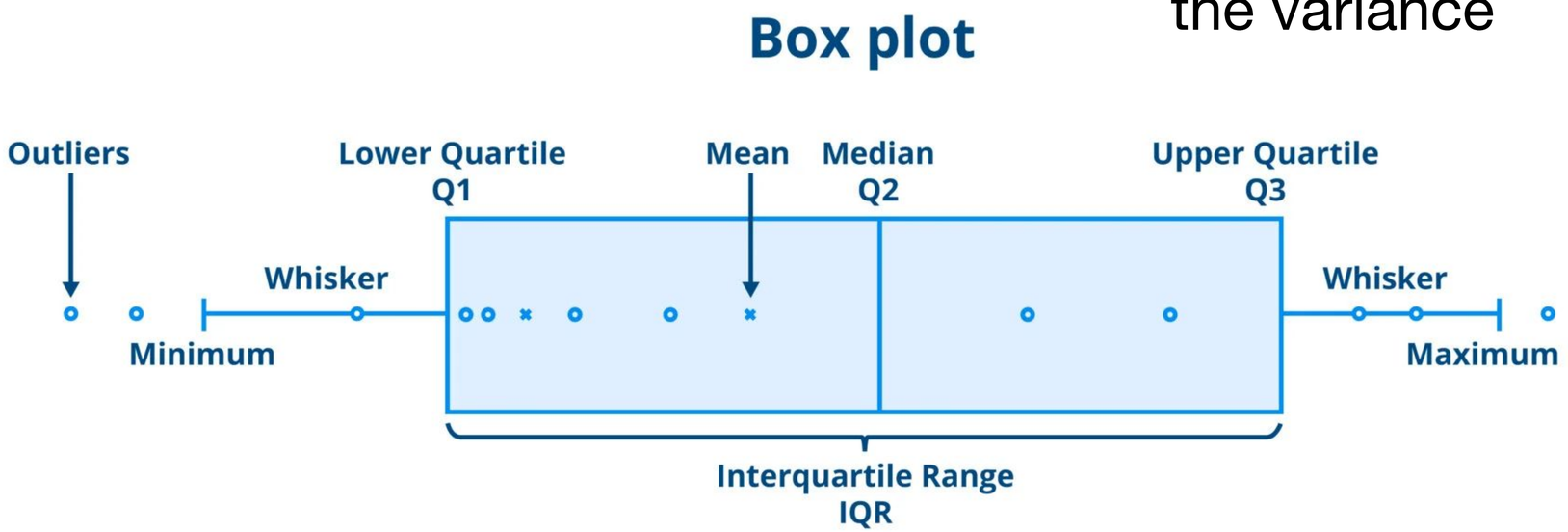
- Detecting outliers or weird values very important, because they can bias results!
- The distribution and spread of individual variables *greatly influences the models and analyses you can use.*
    - **Example:** Statistical tests such as t-tests and ANOVA and models like linear regression require that the variables (or residual errors) are **normally-distributed**

Step 2: Univariate EDA (Quantitative)

Basic descriptive statistics

- Mean: average value of a variable
- Median: middle value of a variable
- Mode: most common value of a variable
- Range: max value - min value

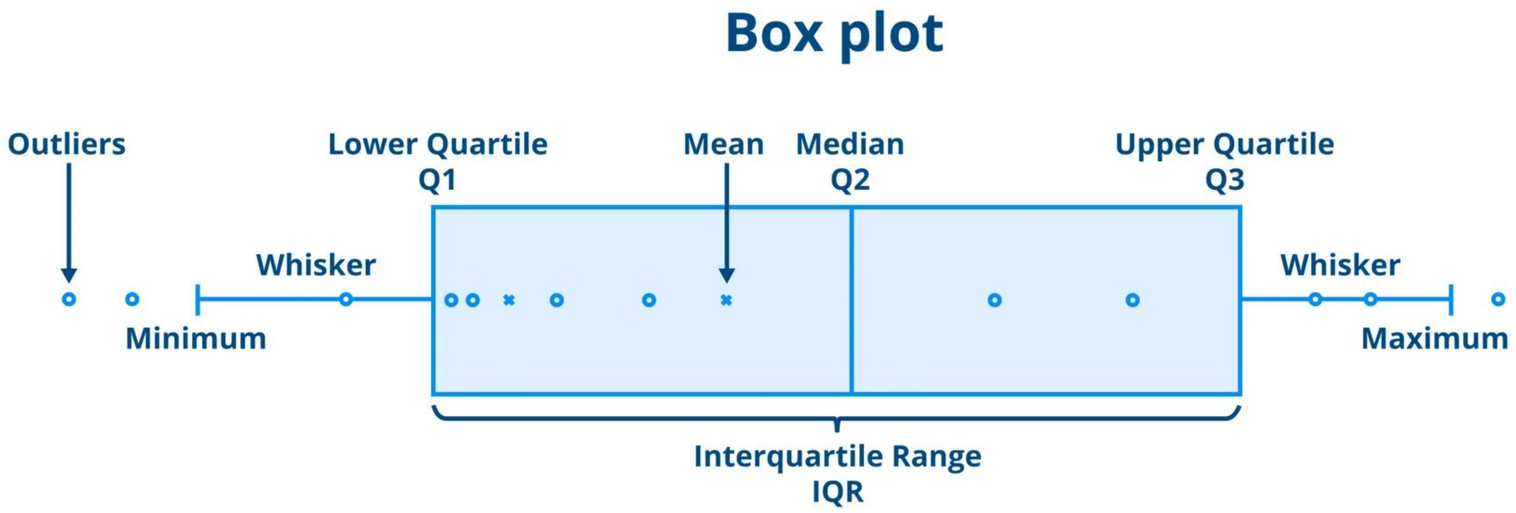
- Interquartile Range (IQR): 75th percentile of variable - 25th percentile of variable
- Variance: how spread out values are around the mean (the average of the squared deviations)
- Standard deviation (SD): square-root of the variance

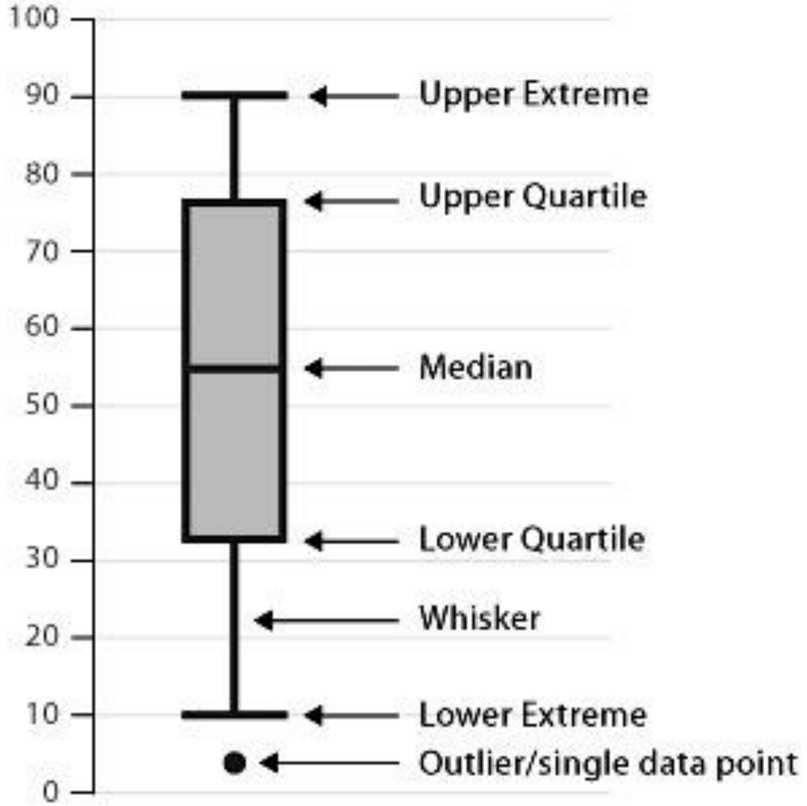


Step 2: Univariate EDA (Quantitative)

EDA for quantitative variables

- Basic summary: mean/median, standard deviation, inter-quartile range, min/max
  - ***Visualization:*** Box plots are great for visualizing descriptive statistics



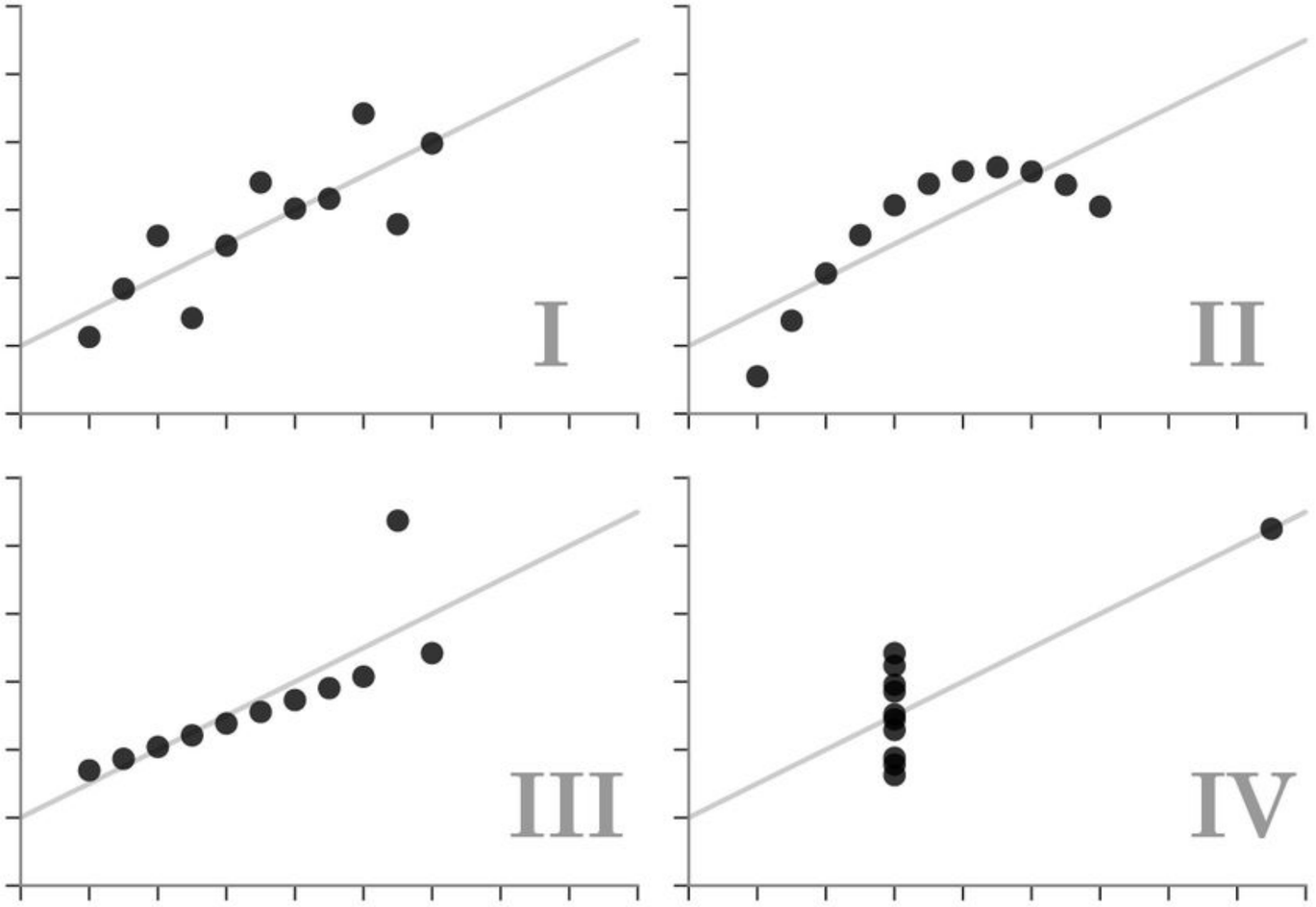


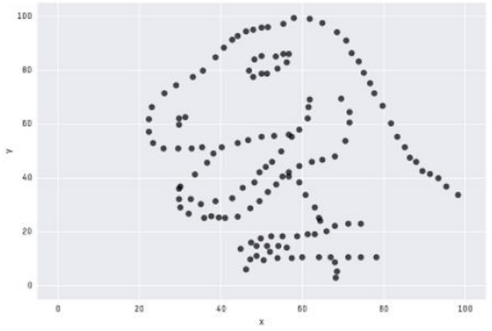
Step 2: Univariate EDA (Quantitative)

But summary statistics alone are NOT enough!

Anscombe's Quartet

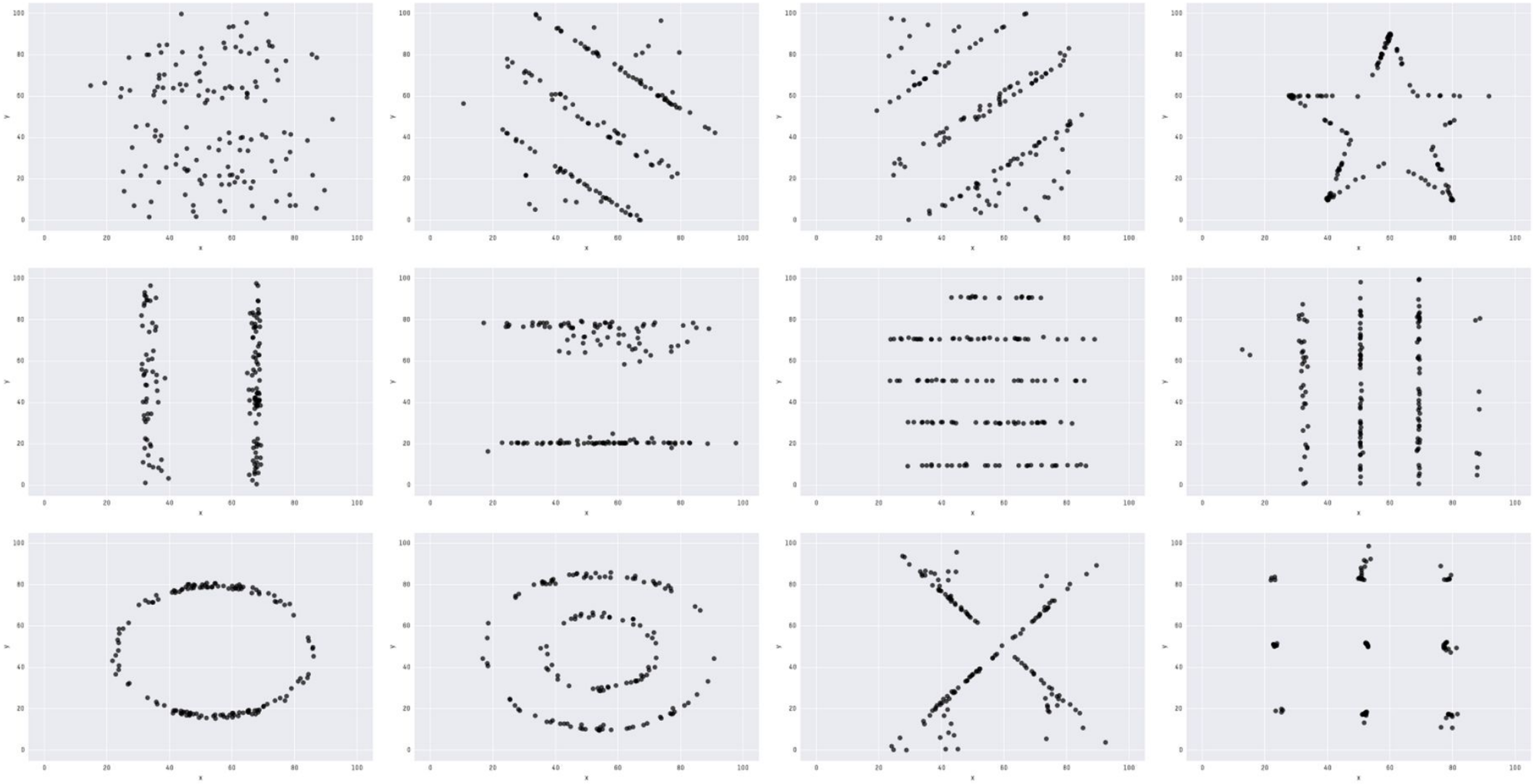
Each dataset has the same summary statistics (mean, standard deviation, correlation), and the datasets are *clearly different*, and *visually distinct*.





X Mean: 54.26
Y Mean: 47.83
X SD : 16.76
Y SD : 26.93
Corr. : -0.06

These datasets all have the **same descriptive statistics!**



Step 2: Univariate EDA (Quantitative)

EDA for quantitative variables

- Basic summary: mean/median, standard deviation, inter-quartile range, min/max
  - *Visualization:* Box plots are great for visualizing descriptive statistics
- Any outliers or weird values (mistakes)?
  - ***Visualization:*** Both box plots and histograms are good for a quick check

	mean	std	min	25%	50%	75%	max
What is your height in inches?	122.33	630.12	5.1	64.0	67.0	70.0	7283

Step 2: Univariate EDA (Quantitative)

EDA for quantitative variables

- Basic summary: mean/median, standard deviation, inter-quartile range, min/max
  - *Visualization:* Box plots are great for visualizing descriptive statistics
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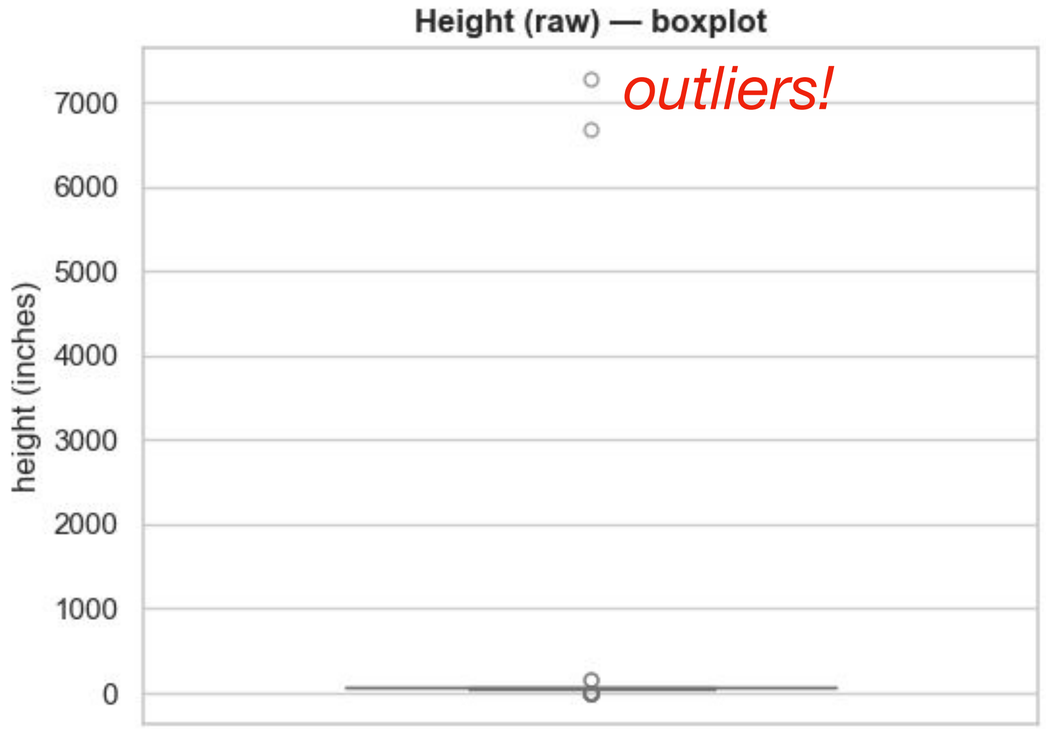
606ft ??

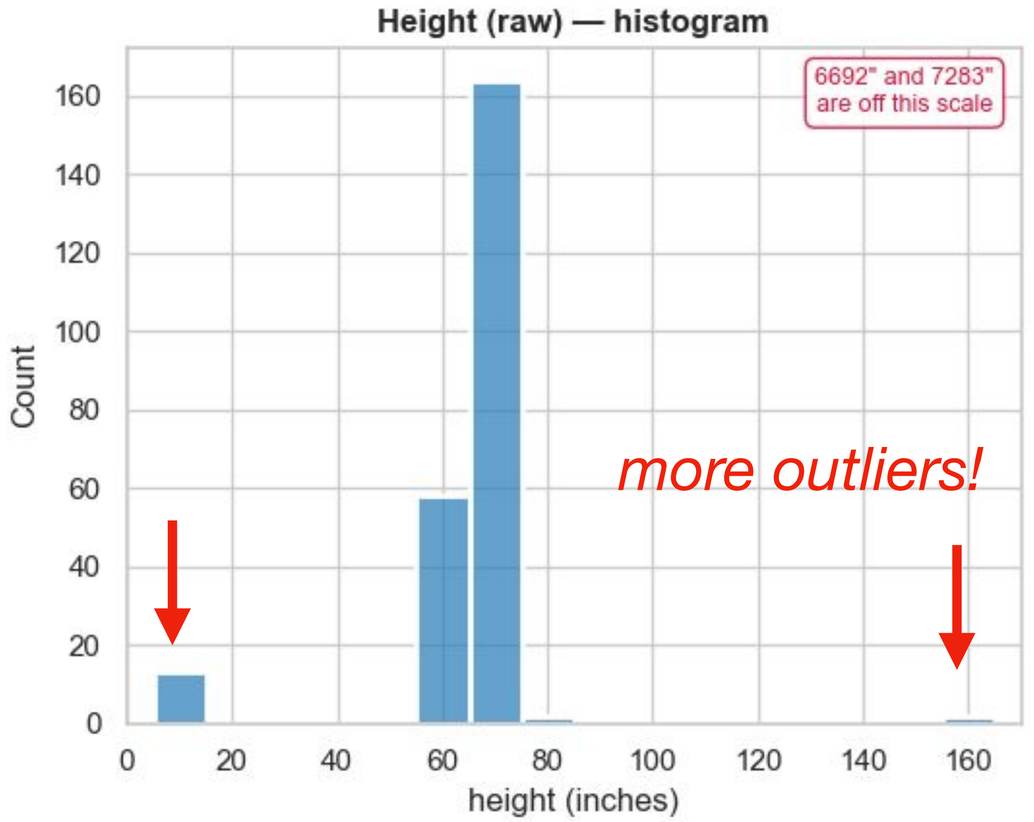
What do you think went wrong?

Step 2: Univariate EDA (Quantitative)

606ft??

	mean	std	min	25%	50%	75%	max
What is your height in inches?	122.33	630.12	5.1	64.0	67.0	70.0	7283





Step 2: Univariate EDA (Quantitative)

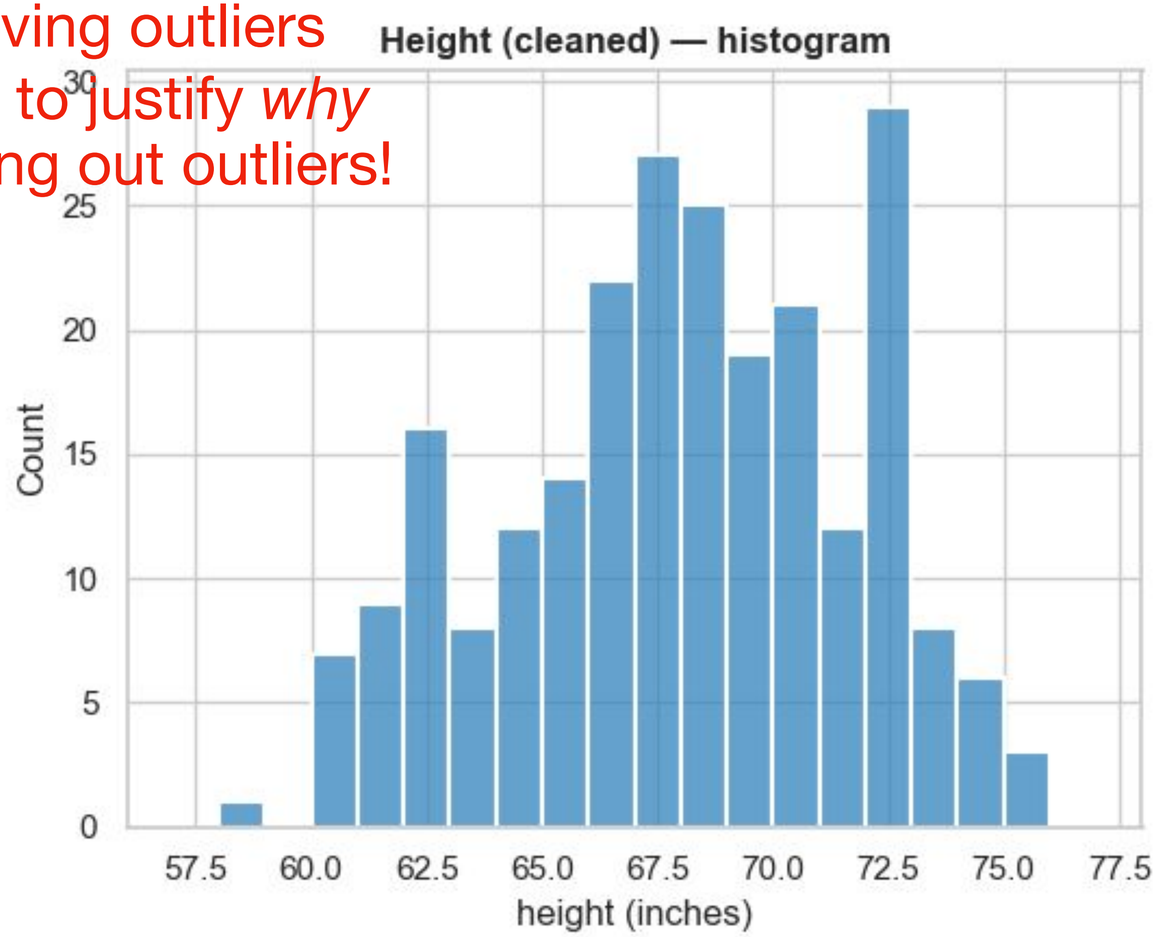
6ft 4in!

	mean	std	min	25%	50%	75%	max
What is your height in inches?	67.6	3.77	58	65.0	67.0	70.0	**76**

after removing outliers

Important to justify *why* you're taking out outliers!





Step 2: Univariate EDA (Quantitative)

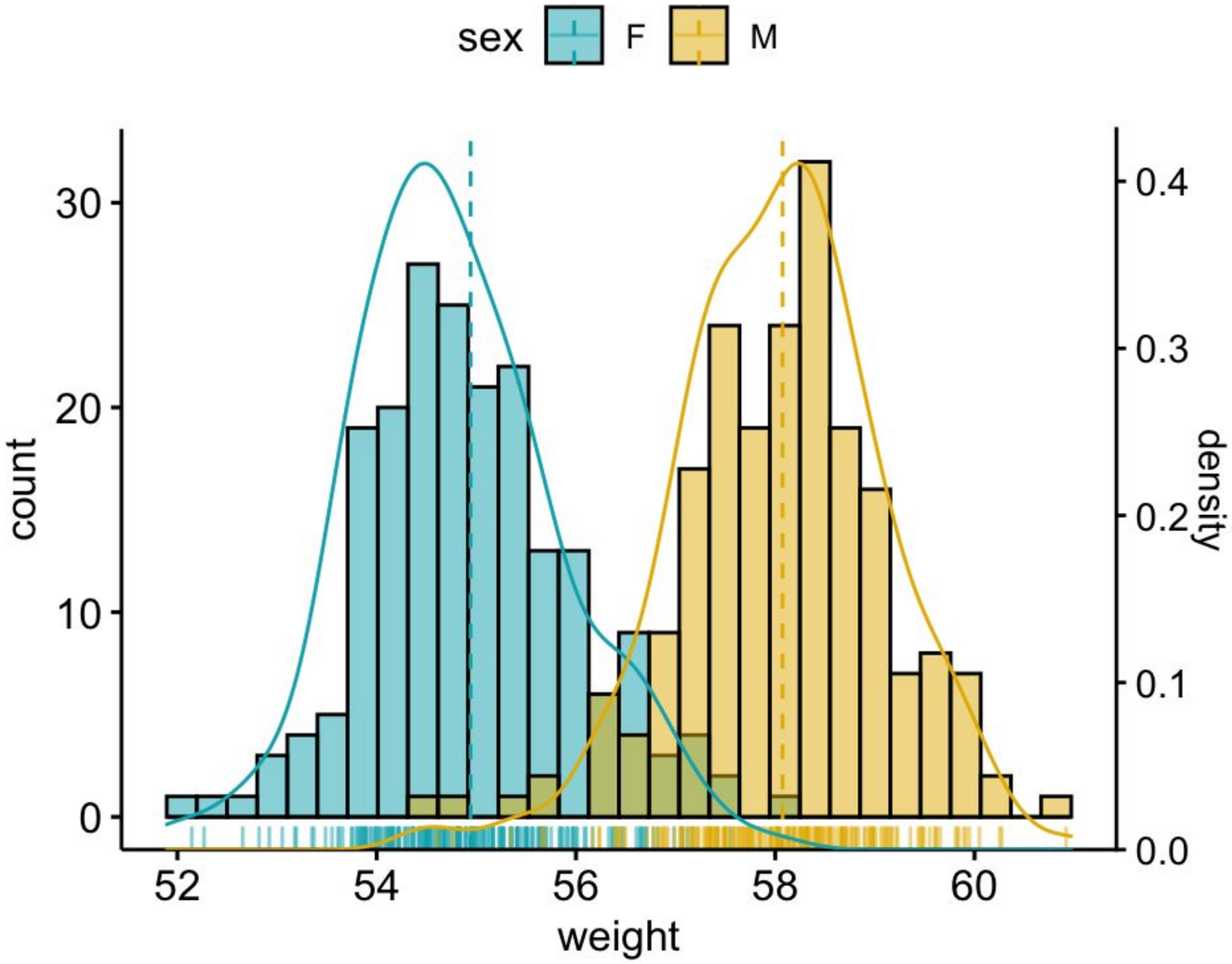
EDA for quantitative variables

- Basic summary: mean/median, standard deviation, inter-quartile range, min/max
  - ***Visualization:*** Box plots are great for visualizing descriptive statistics
- Any outliers or weird values (mistakes)?
  - ***Visualization:*** Both box plots and histograms are good for a quick check
- What is the variable's distribution? Skewness? Multi-modal?
  - ***Visualization:*** Histogram or density plots

Step 2: Univariate EDA (Quantitative)

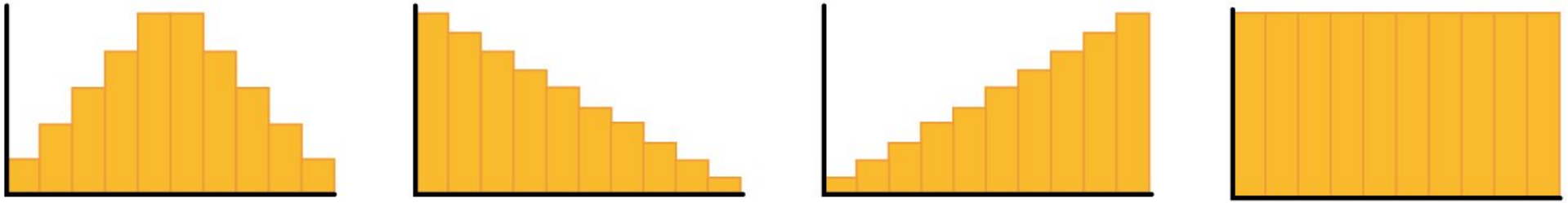
Histogram vs. density plot

- Both show distribution of a variable
- **Histogram** shows the counts (how many) in each "bin"
- **Density plot** is a probability density function (PDF)
  - Y-axis is density (not probability)
  - Density: the area between 2 points on x-axis is the probability of data falling in that range



Step 2: Univariate EDA (Quantitative)

Symmetric (normal) vs skewed and uniform distriutions



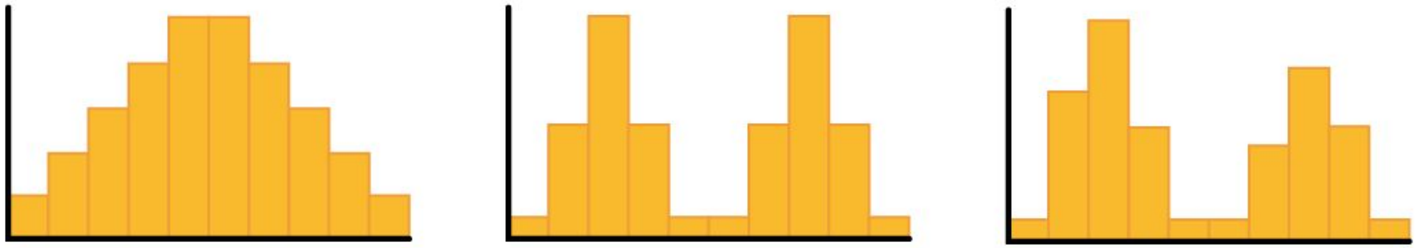
Normal distribution
(unimodal, symmetric,
the “bell curve”)

Right-skewed distribution
(Positively-skewed)

Left-skewed distribution
(Negatively-skewed)

Uniform distribution
(equal spread,
no peaks)

Unimodal vs bimodal distributions



Normal distribution
(unimodal, symmetric,
the “bell curve”)

Symmetric bimodal distribution
(two modes)

Non-symmetric bimodal distribution
(two modes)

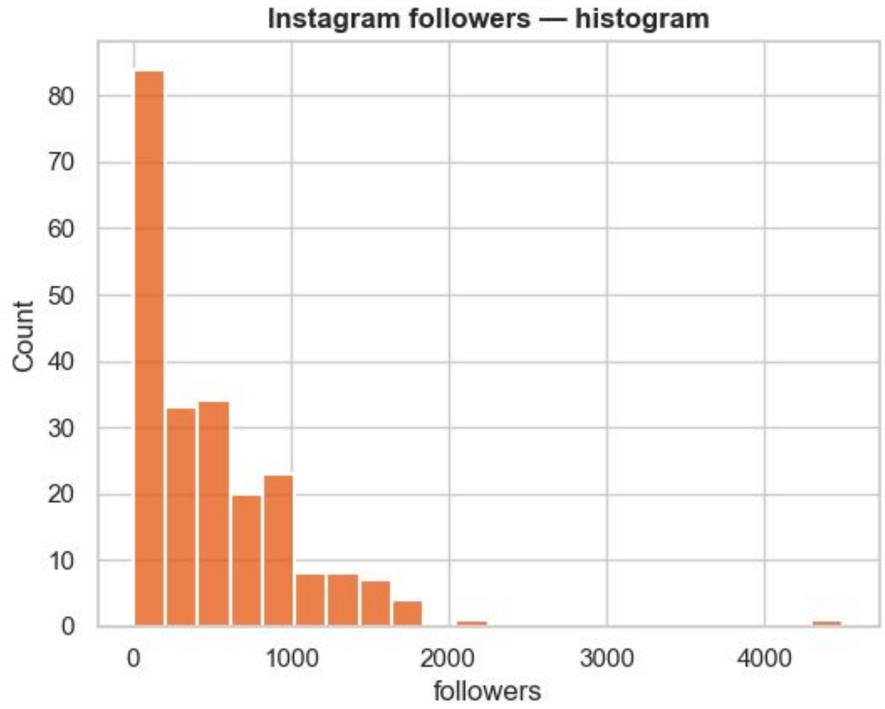
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course code:
WSTAPX

The number of Instagram followers among COGS 9 students is…



(A) Right-skewed

(B) Left-skewed

(C) Uniform

(D) Normally-distributed

(E) I’m not sure / I’m confused

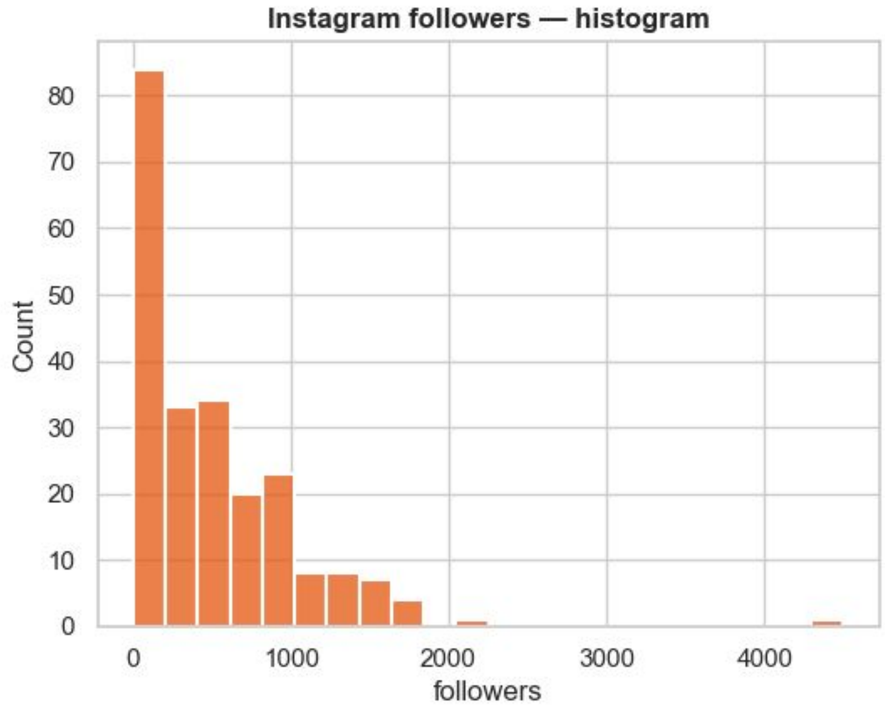
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course code:
WSTAPX

The number of instagram followers among COGS 9 students is…



(A) Right-skewed

(B) Left-skewed

(C) Uniform

(D) Normally-distributed

(E) I'm not sure / I'm confused

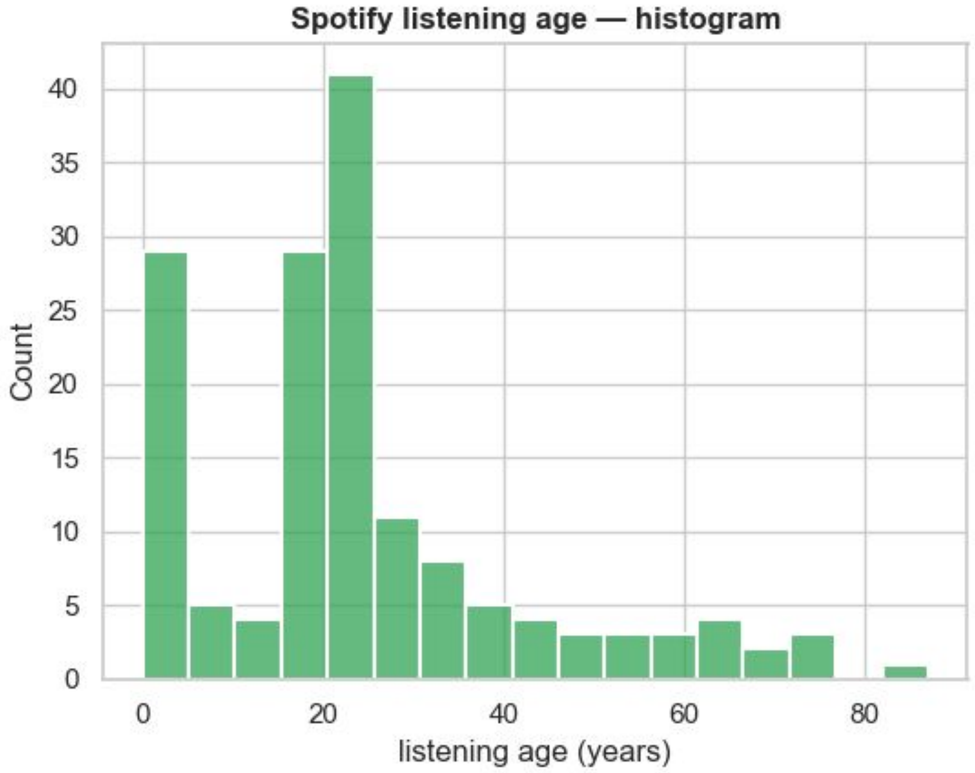
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course code:
WSTAPX

Spotify listening age among COGS 9 students is…



(A) Uniform

(B) Bi-modal

(C) Tri-modal

(D) Normally-distributed

(E) I'm not sure / I'm confused

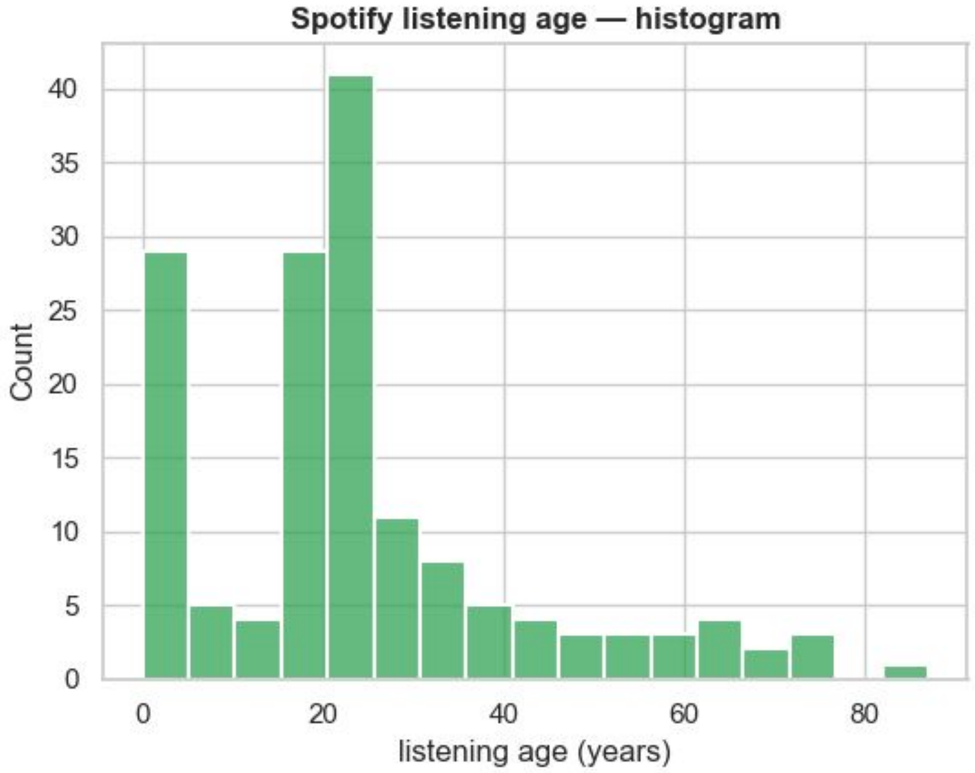
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course code:
WSTAPX

Spotify listening age among COGS 9 students is…



(A) Uniform

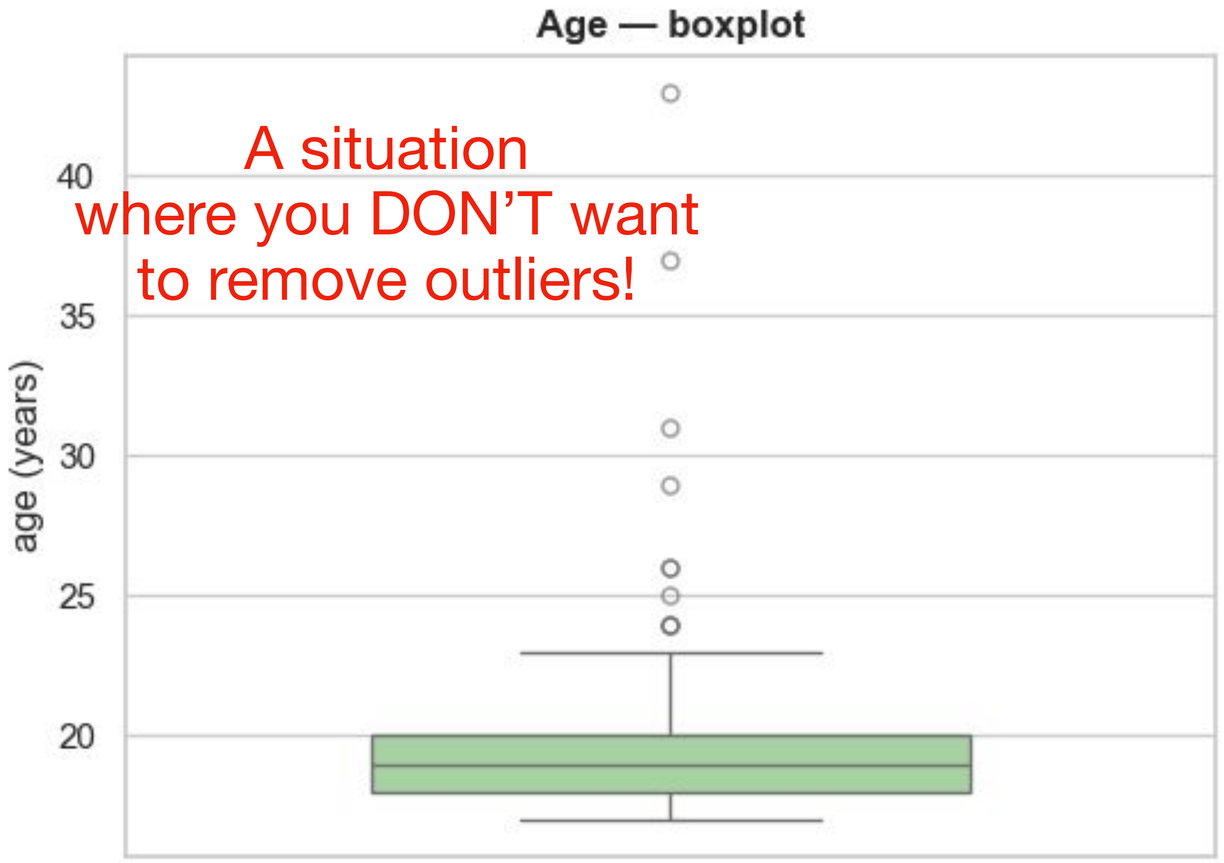
(B) Bi-modal

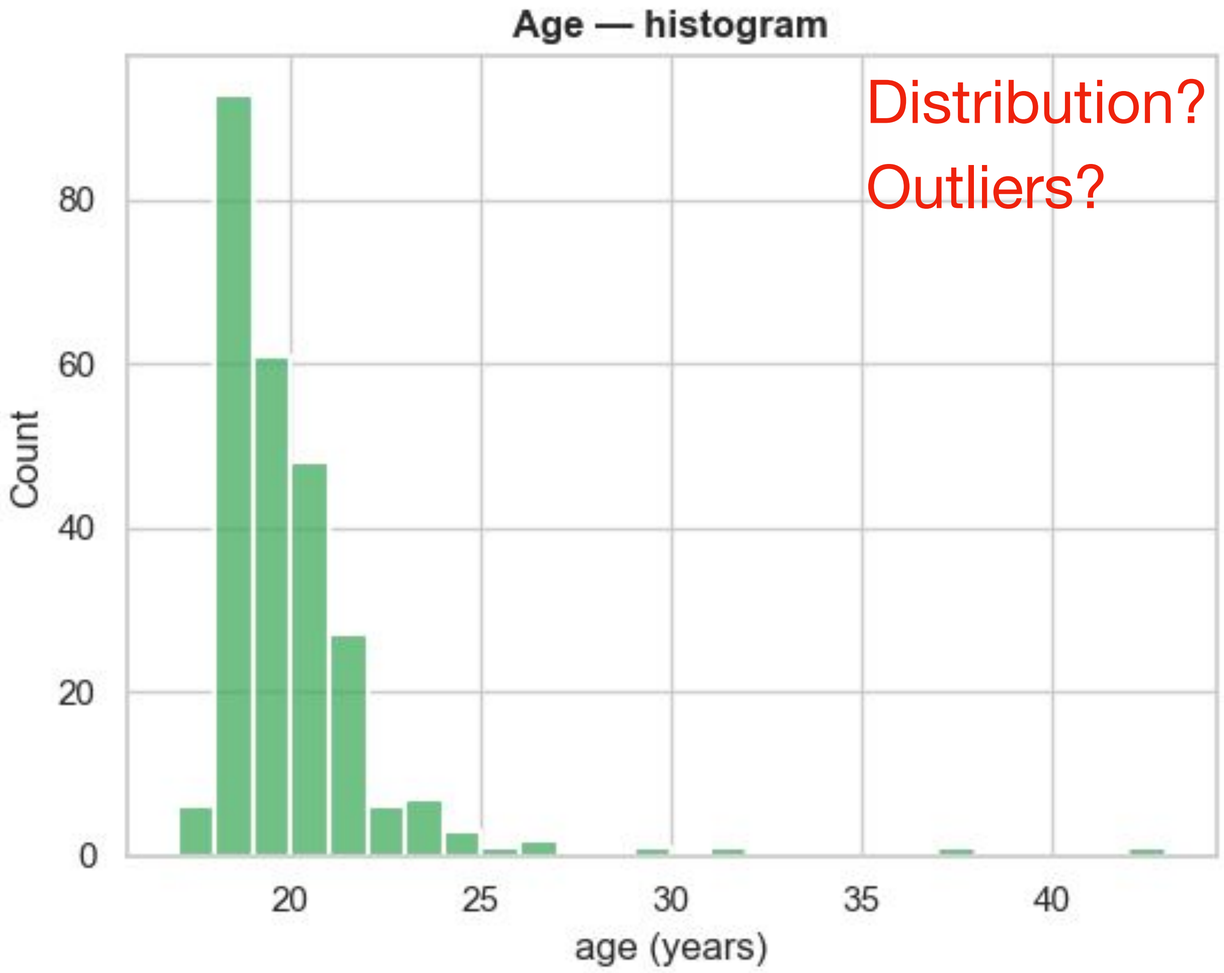
(C) Tri-modal

(D) Normally-distributed

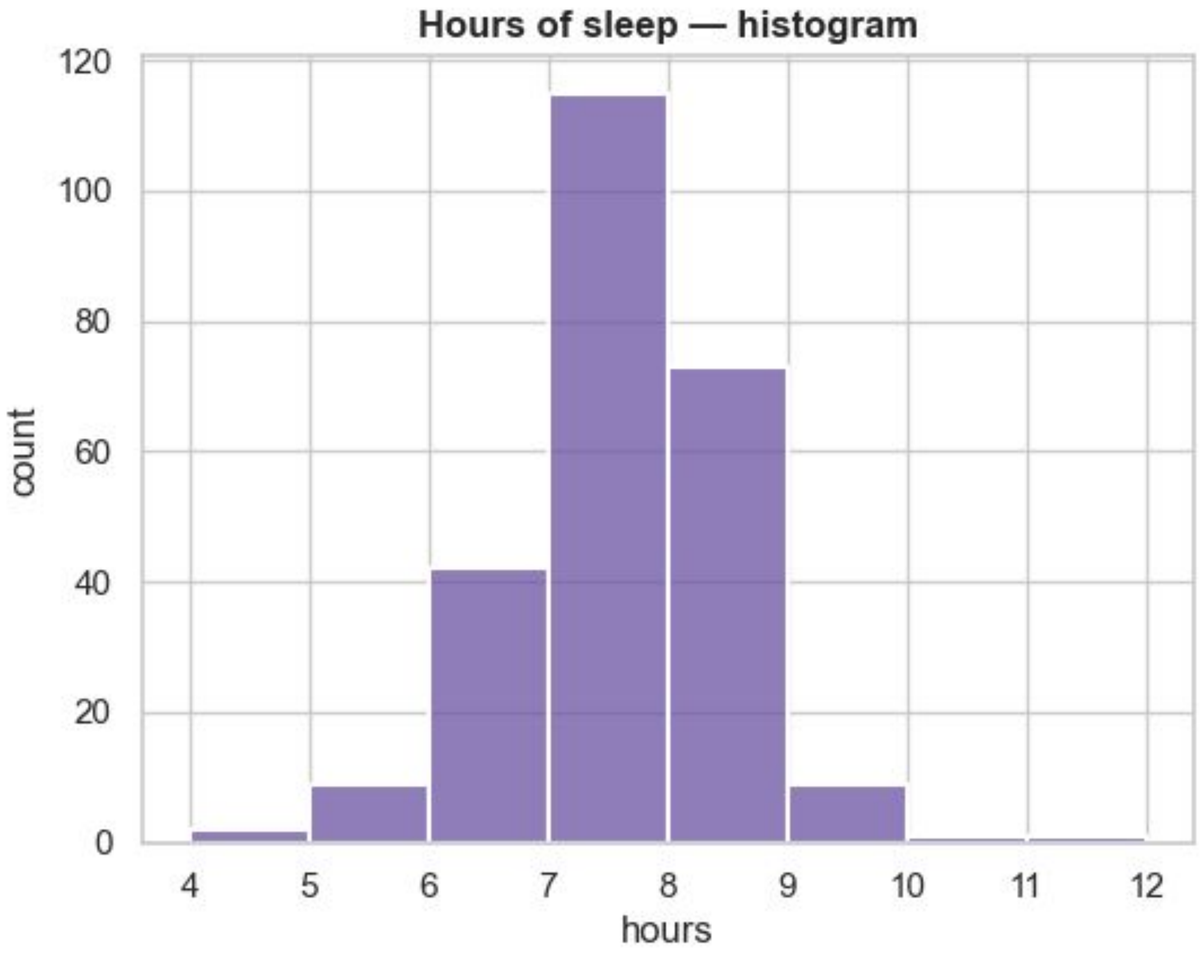
(E) I'm not sure / I'm confused

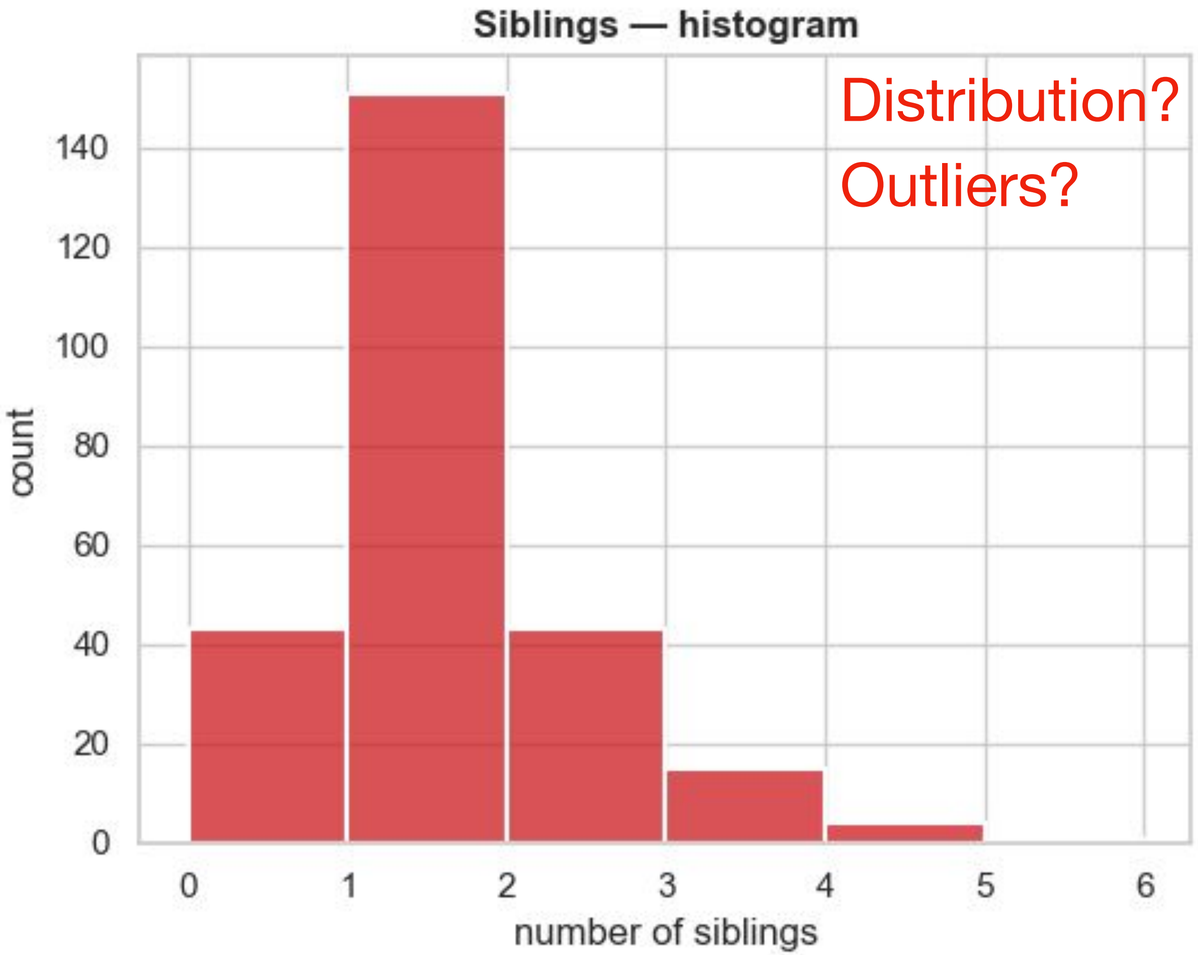
Step 2: Univariate EDA (Quantitative)





Step 2: Univariate EDA (Quantitative)





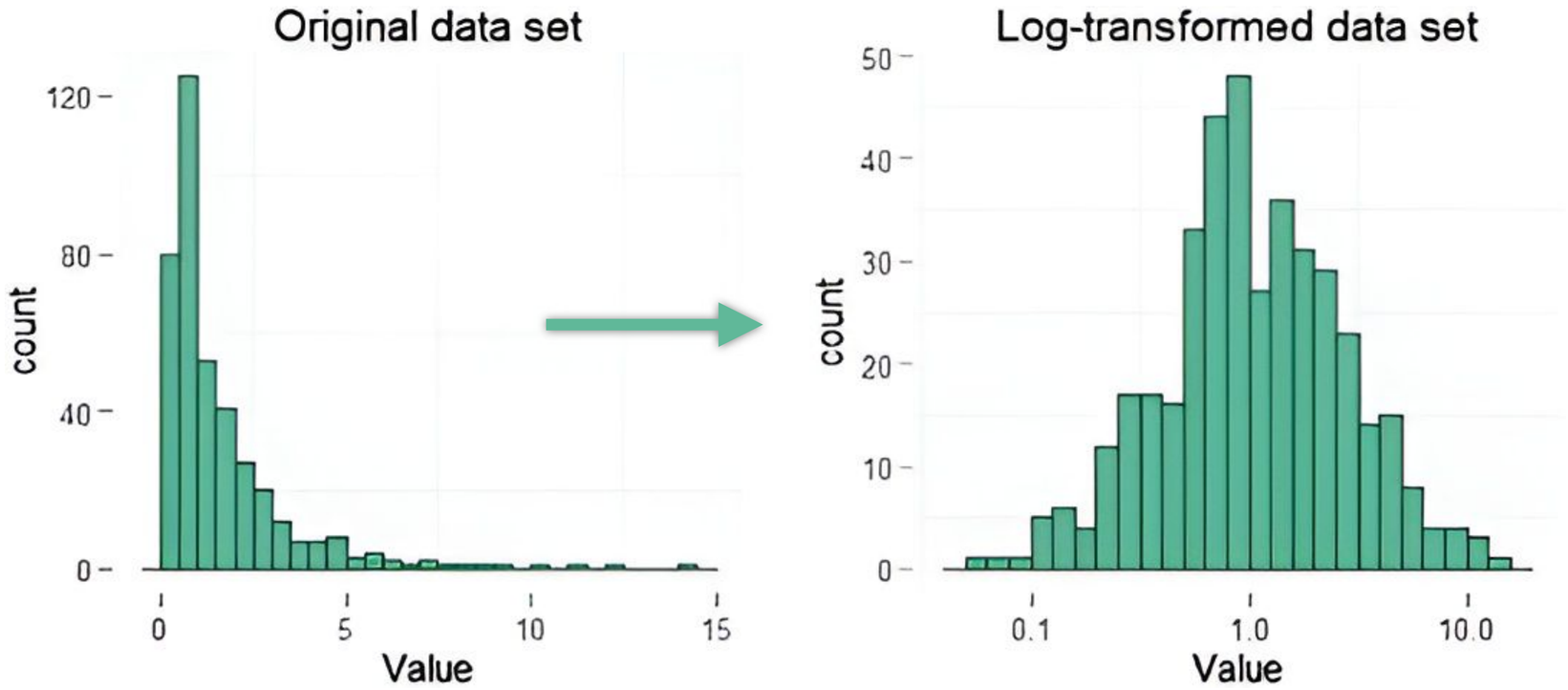
Distribution?

Outliers?

Step 2: Univariate EDA (Quantitative)

Why is this important?:

- The distribution and spread of individual variables *greatly influences the models and analyses you can use.*
  - **Example:** T-tests requires variables be **normally-distributed**

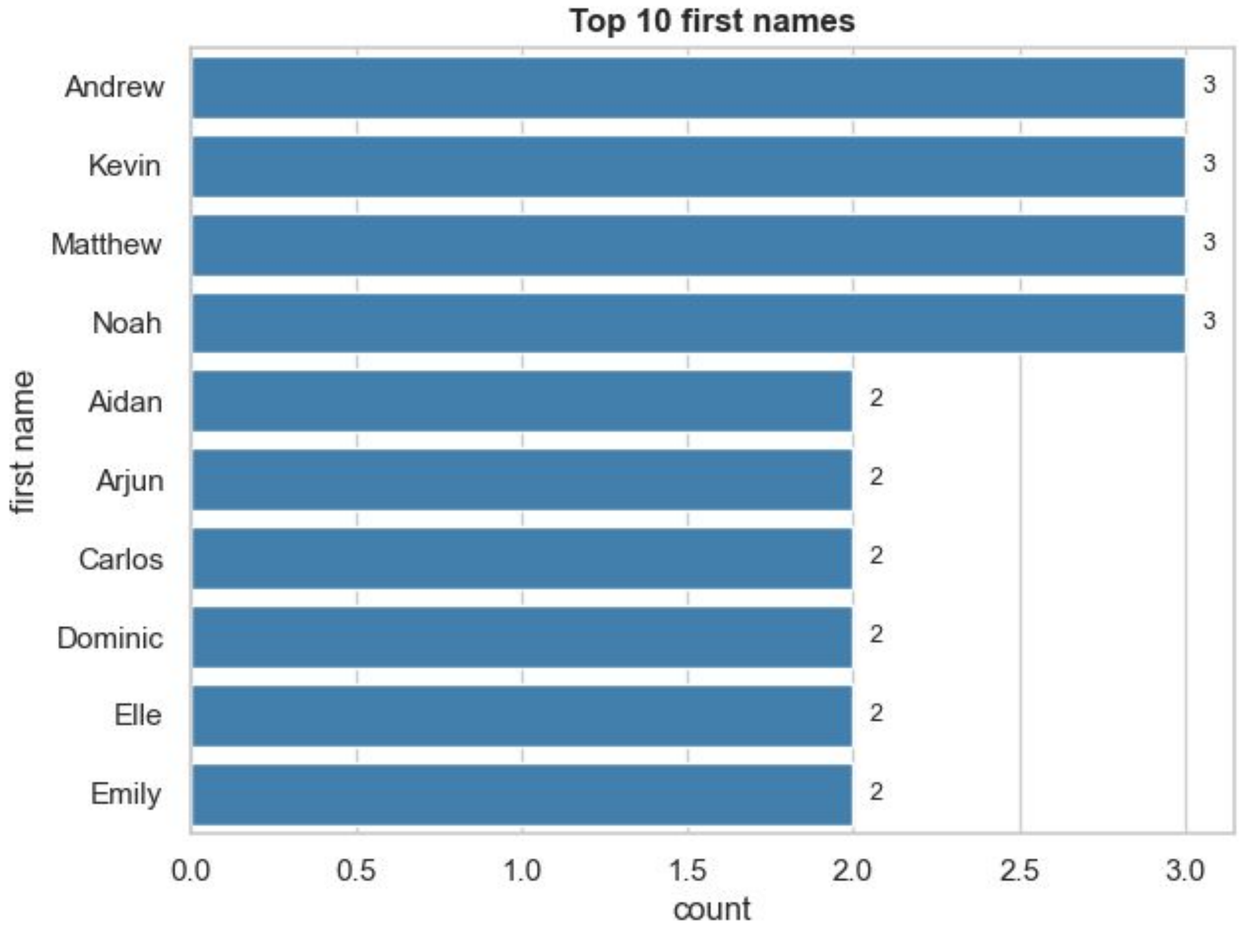


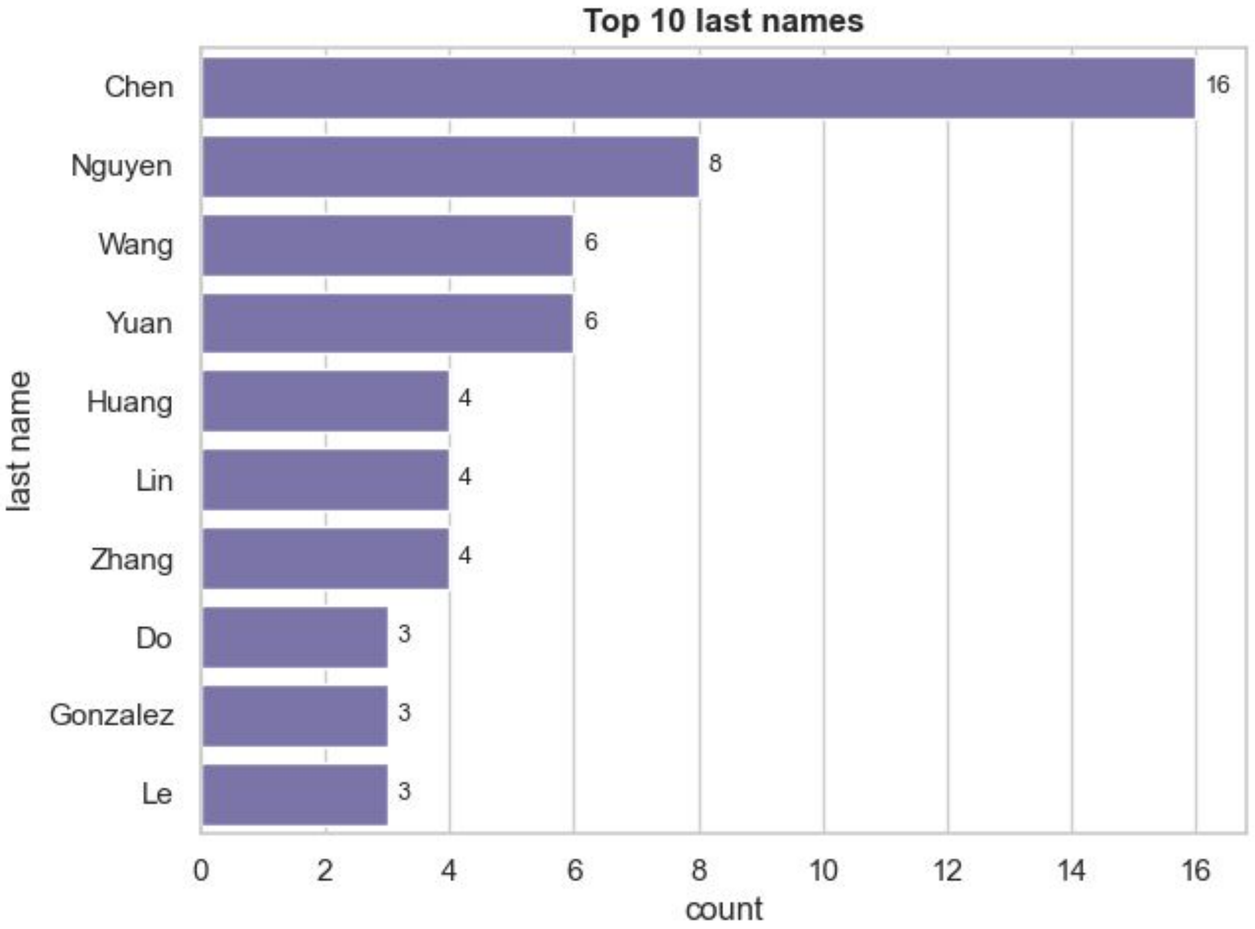
Step 2: Univariate EDA (Qualitative)

EDA for qualitative variables

- Basic summary: categorical counts for nominal or ordinal data
  - Are the data biased in any way?
  - Class imbalance: more samples of one category than another
  - ***Visualization:*** bar plots, pie charts, frequency tables

Step 2: Univariate EDA (Qualitative)



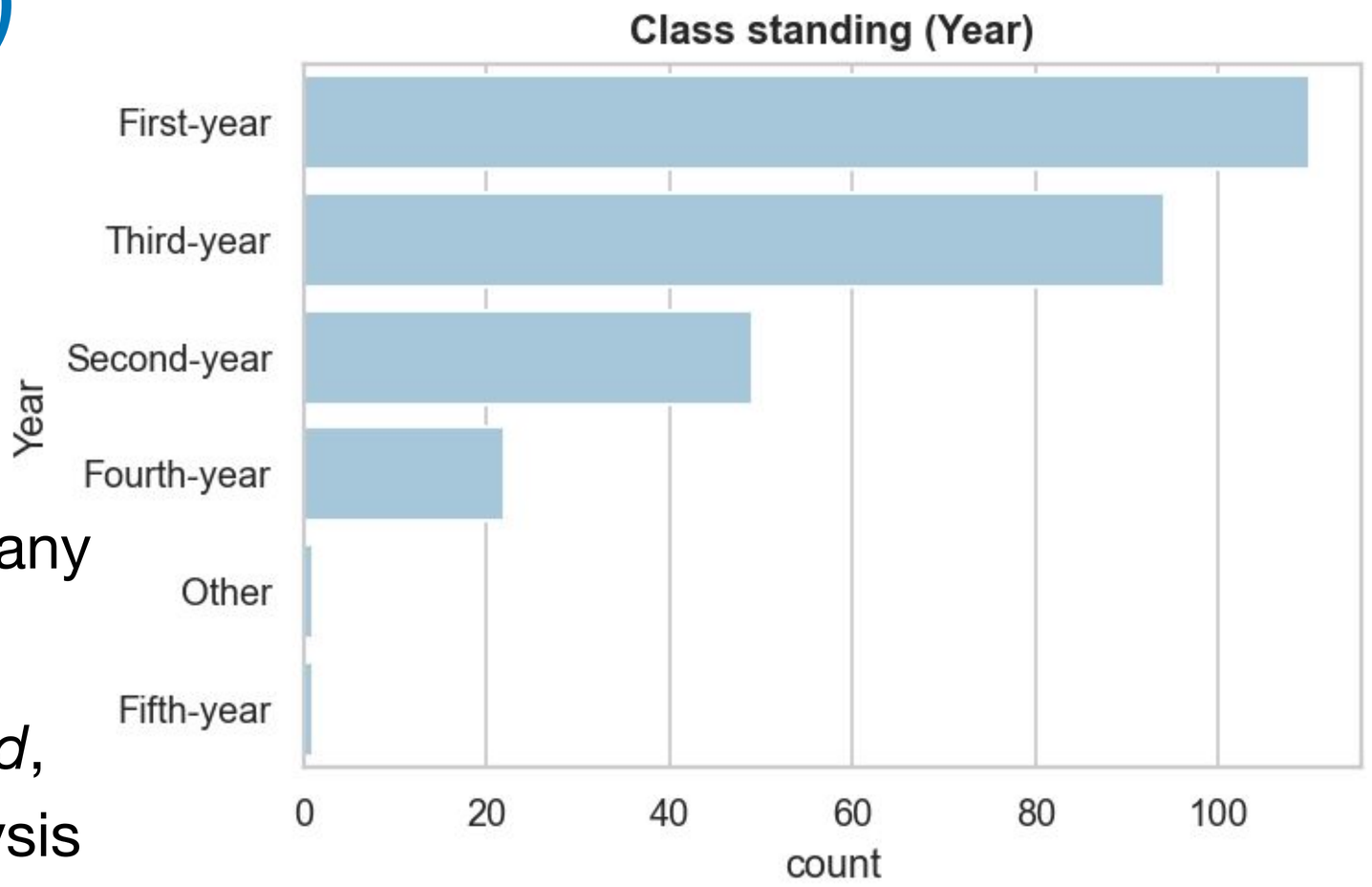


Discussion (~2 mins)

Consider the following question:

Do first-years get more sleep than fourth-years?

1. Why is it important to first do univariate EDA to know how many students are in each category?
2. If the categories are *imbalanced*, how does this affect data analysis and interpretation?



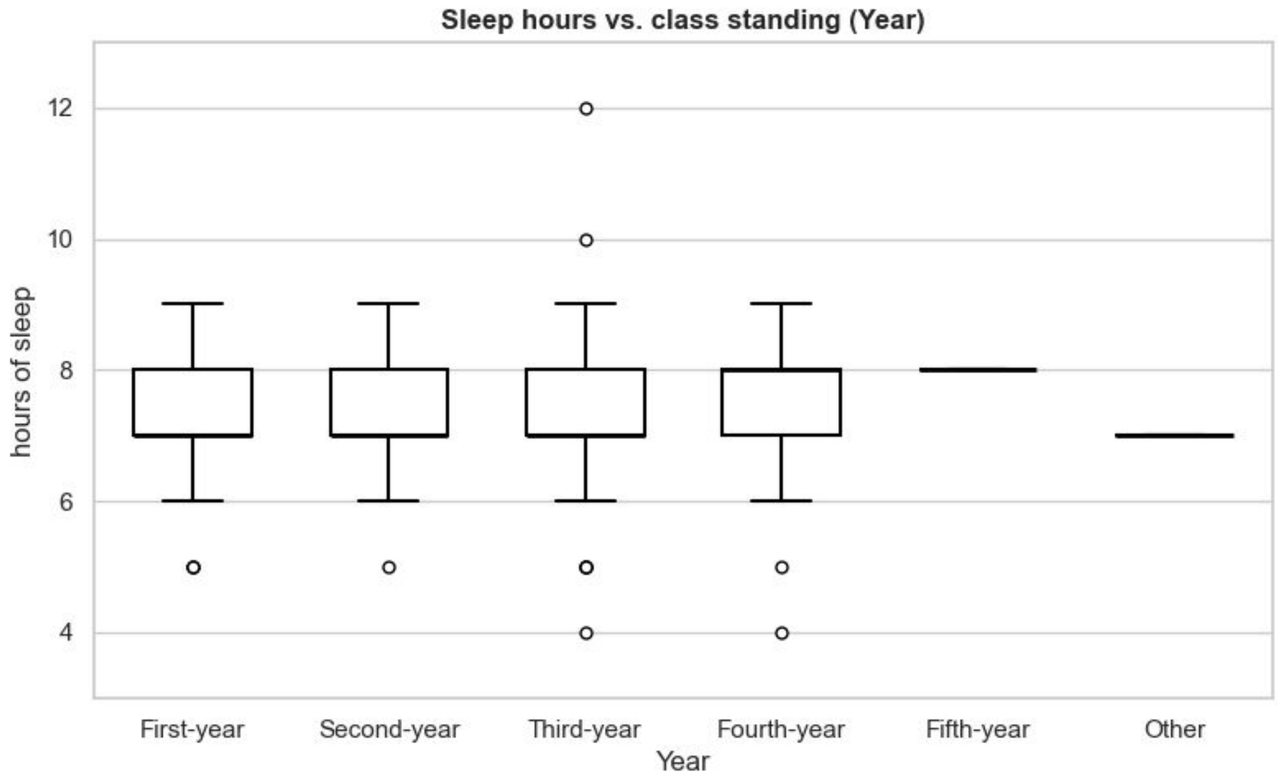
Discussion (~2 mins)

Consider the following question:

Do first-years get more sleep than fourth-years?

Consequences of class imbalance

- Small *n* reduces **statistical power**
- **Cannot make meaningful conclusions:** very few fifth-years, so the mean/median sleep can swing a lot due to just 1–2 students
- Comparison is not equal!



Step 2: Univariate EDA (Qualitative)

EDA for qualitative variables

- Basic summary: categorical counts for nominal or ordinal data
  - Are the data biased in any way?
  - Class imbalance: more samples of one category than another
  - *Visualization:* bar plots, pie charts, frequency tables
- **Pre-processing unstructured data** (e.g., text, images, speech)
  - Text analysis: Counting key words
  - Image data: Detecting objects in photos, labeling photos (computer vision)
  - Audio analysis: Detecting and categorizing sounds or words from waveforms
  - ***Visualizations:*** depends highly on the data type!

```python
import re
from collections import Counter

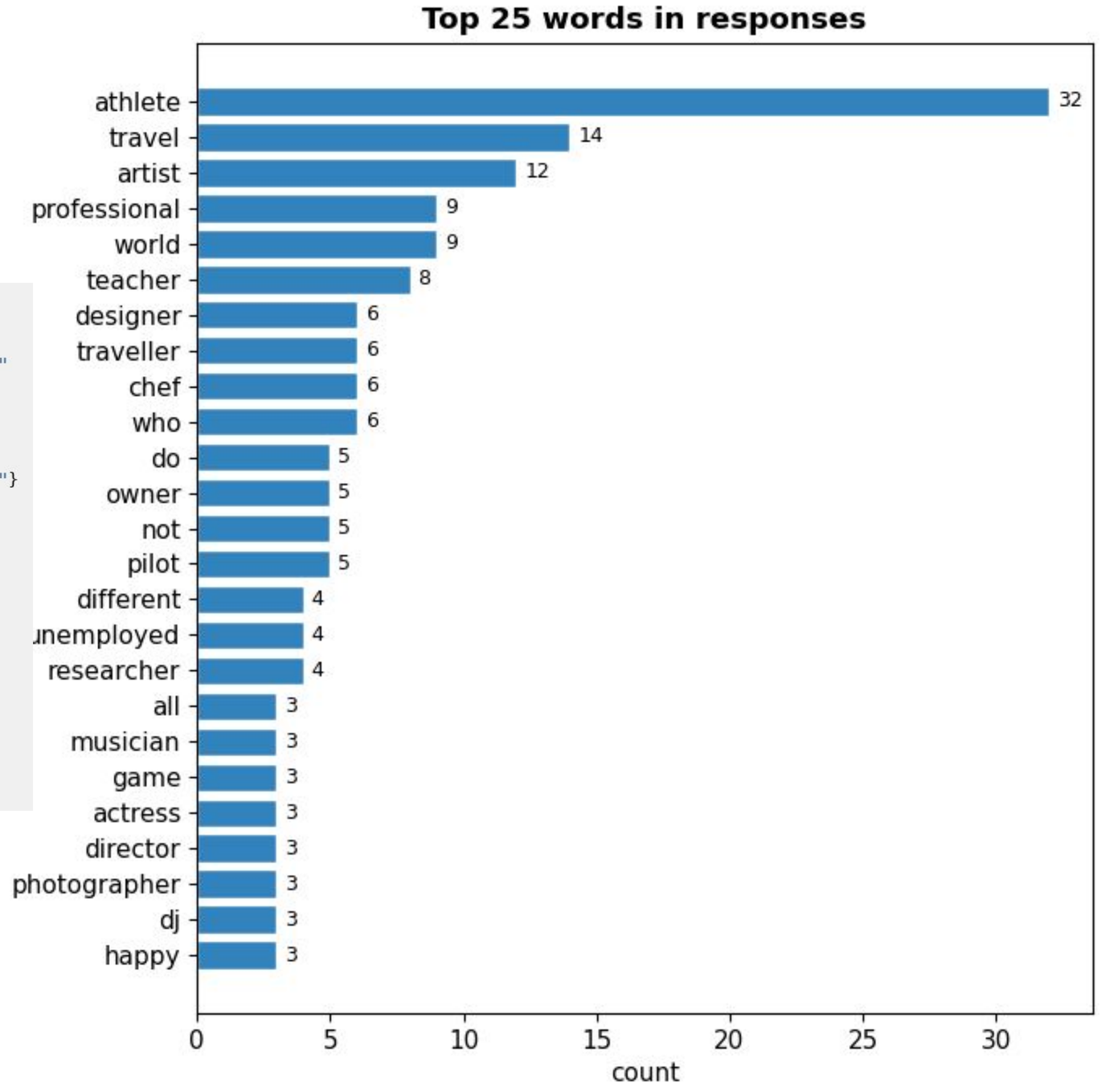
col = "If money wasn't an object, what would you want to be when you grow up?"
s = df_clean[col].dropna().astype(str)

stopwords = {"a", "the", "and", "or", "if", "so", "to", "of", "in",
             "on", "for", "with", "is", "are", "i", "me", "my", "we",
             "you", "your", "it", "its", "this", "that", "would", "want",
             "like", "really", "just", "what", "grow", "up", "money", "object"}

alias = {   # merge synonyms into one canonical word
    "basketball": "athlete", "golfer": "athlete", "swimmer": "athlete",
    "runner": "athlete", "soccer": "athlete", "nba": "athlete",
    "traveler": "travel", "traveling": "travel", "travels": "travel",
}

def tokenize(text):
    text = re.sub(r"[^a-z\s]", " ", text.lower())
    tokens = [t for t in text.split()
              if len(t) >= 2 and t not in stopwords]
    return [alias.get(t, t) for t in tokens]

tokens = [tok for resp in s for tok in tokenize(resp)]
counts = Counter(tokens)
top = counts.most_common(25)          # raw word counts (not TF-IDF)
```



Next time!

Bivariate and multivariate EDA!

Statistical Inference, Hypothesis testing!